



Lab Report: Diffraction Grating Spectrometry

Student Name: _____

Date: _____

1. Objective

To measure the wavelength of light by observing the diffraction angles through a grating.

2. Data Collection

$N = \text{lines/mm}$, $d = 1/N \times 10^{-3} \text{ m}$.

Source Color	Lines/mm (N)	Grating d	Angle θ ($m = 1$)	Calc. λ
Green	300			
Green	600			
Unknown	600			

3. Analysis Questions

1. Calculate the wavelength for the "Unknown" source using the grating equation $d \sin \theta = \lambda$.

2. Why are the spots in a grating pattern much sharper than the fringes in a double-slit pattern?